

BERLIN TEGEL, Germany

WMO: 103820

Lat: 52.5644N

Lon: 13.3089E

Elev: 37

StdP: 100.88

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.2	-14.9	1.0	-8.9	-11.9	1.4	-5.6	11.7	7.3	10.8	6.5	3.8	70	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	31.2	19.2	29.2	18.6	27.2	18.1	20.7	27.6	20.0	26.6	19.2	25.3	3.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	22.6	17.9	12.9	22.4	17.0	12.2	21.7	59.7	27.6	57.2	26.9	54.5	24.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.4	7.5	-12.9	34.8	3.8	2.0	-15.6	36.2	-17.8	37.4	-19.9	38.5	-22.7	39.9		
(5)			-13.3	22.8	3.7	0.8	-16.0	23.4	-18.2	23.9	-20.3	24.4	-23.0	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	0.8	1.6	5.2	10.8	14.5	18.3	20.3	19.4	15.7	10.7	6.1	3.0	(6)
(7)		DBStd	7.88	4.86	4.21	4.36	4.19	4.23	3.78	3.52	3.25	3.26	3.57	3.69	4.29	(7)
(8)		HDD10.0	1101	286	236	157	39	7	0	0	0	1	36	123	217	(8)
(9)		HDD18.3	3041	544	469	408	226	132	47	19	24	93	238	366	475	(9)
(10)		CDD10.0	1317	0	0	7	64	148	248	321	293	172	57	8	1	(10)
(11)		CDD18.3	214	0	0	0	1	15	44	81	58	14	1	0	0	(11)
(12)		CDH23.3	1843	0	0	0	15	141	404	728	454	99	3	0	0	(12)
(13)		CDH26.7	573	0	0	0	2	34	128	251	142	16	0	0	0	(13)
(14)	Wind	WSAvg	3.6	4.3	3.9	3.9	3.6	3.4	3.3	3.2	3.1	3.2	3.3	3.7	4.1	(14)
(15)	Precipitation	PrecAvg	613	47	33	34	36	62	63	70	66	43	44	49	47	(15)
(16)		PrecMax	907	110	74	96	82	148	181	202	160	106	137	99	130	(16)
(17)		PrecMin	430	15	3	11	3	8	2	8	8	5	5	7	11	(17)
(18)		PrecStd	126	23	20	19	22	33	40	45	32	22	30	22	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.8	19.1	25.1	30.1	33.0	34.9	33.2	29.0	22.9	16.2	13.1	(19)
(20)			MCWB	9.1	7.4	10.9	14.3	18.1	20.2	20.0	20.1	18.3	15.9	13.7	10.2	(20)
(21)		2%	DB	10.2	10.2	16.0	22.8	27.0	30.0	31.8	30.2	26.1	20.0	14.1	11.1	(21)
(22)			MCWB	8.4	7.4	9.9	13.4	16.9	18.6	19.2	19.7	17.5	14.7	11.7	9.1	(22)
(23)		5%	DB	8.9	9.0	13.8	20.9	24.2	27.9	29.8	27.9	24.0	18.0	12.8	10.0	(23)
(24)			MCWB	7.4	6.7	8.9	12.4	15.3	18.1	18.9	18.6	16.6	13.8	10.4	8.2	(24)
(25)		10%	DB	7.2	7.2	11.8	18.8	22.2	25.2	27.2	25.8	21.8	16.1	11.1	8.8	(25)
(26)			MCWB	5.9	5.0	7.9	11.7	14.5	17.1	18.2	17.7	15.7	13.1	9.2	7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.4	12.3	15.4	19.5	21.7	21.6	21.7	19.8	16.8	13.7	11.1	(27)
(28)			MCDWB	11.2	10.1	16.9	23.1	26.4	29.2	28.2	28.1	26.5	20.9	15.7	12.9	(28)
(29)		2%	WB	8.7	7.7	10.7	14.2	18.1	20.1	20.7	20.6	18.3	15.4	11.9	9.6	(29)
(30)			MCDWB	10.2	10.0	14.5	20.5	25.0	26.7	27.7	27.4	23.8	19.0	13.8	10.7	(30)
(31)		5%	WB	7.2	6.5	9.5	13.2	16.8	19.1	20.0	19.7	17.2	14.2	10.6	8.4	(31)
(32)			MCDWB	8.3	8.7	12.8	19.1	22.5	25.4	26.7	25.7	22.1	17.3	12.3	9.8	(32)
(33)		10%	WB	5.9	5.1	8.2	12.1	15.5	18.0	19.3	18.7	16.2	13.2	9.7	7.1	(33)
(34)			MCDWB	7.0	7.0	11.2	17.9	20.7	23.9	25.7	23.7	20.8	15.8	11.0	8.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.3	5.5	7.6	9.9	9.9	9.7	9.4	9.4	8.9	7.1	4.9	4.2	(35)
(36)			MCDBR	5.3	7.8	11.6	12.8	12.5	12.9	12.9	12.5	12.0	9.5	6.1	5.2	(36)
(37)			MCWBR	4.5	5.6	6.6	6.0	5.4	4.9	4.5	4.8	5.3	5.5	4.4	4.2	(37)
(38)		5% WB	MCDWB	4.8	7.2	10.1	11.7	11.4	11.3	10.9	10.4	10.2	8.7	5.7	5.2	(38)
(39)			MCWBR	4.3	5.5	6.3	6.0	5.4	5.1	4.6	4.9	5.3	5.4	4.4	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.353	0.393	0.419	0.416	0.413	0.434	0.427	0.389	0.376	0.351	0.321	(40)	
(41)		taud	2.310	2.284	2.194	2.199	2.251	2.283	2.237	2.269	2.351	2.348	2.351	2.321	(41)	
(42)		Ebn at Noon	649	743	782	812	836	841	813	796	784	708	615	586	(42)	
(43)		Edh at Noon	63	86	115	129	128	126	129	119	98	81	61	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.62	1.32	2.35	3.94	4.88	5.19	4.86	4.17	3.05	1.68	0.77	0.46	(44)	
(45)		RadStd	0.07	0.23	0.26	0.51	0.58	0.41	0.65	0.33	0.34	0.20	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page